

Can psychedelic-assisted psychotherapy play a role in enhancing motivation to change in addiction treatment settings?

MARK KANG¹, LINDSAY MACKAY^{1,2}, DEVON CHRISTIE¹,
CODY CALLON² and ELENA ARGENTO^{1,2*} 

¹ Department of Medicine, University of British Columbia, Vancouver, Canada

² BC Centre on Substance Use, Vancouver, Canada

Received: July 21, 2021 • Revised manuscript received: February 3, 2022 • Accepted: February 22, 2022

Published online: March 11, 2022

COMMENTARY



ABSTRACT

Despite growing availability of several evidence-based approaches in the treatment of substance use disorders, existing pharmacotherapy and psychosocial interventions continue to have significant limitations, such as low treatment retention rates and high rates of relapse. There is a need to develop new strategies and models to address these limitations and target underlying psychosocial drivers of addiction, such as motivation to change – a crucial factor in achieving positive addiction treatment outcomes. Re-emerging clinical evidence and literature signal the promise of psychedelic-assisted psychotherapies as being novel, adjunctive treatments for a range of mental health and substance use disorders, encouraging further research. However, there remains a lack of formally validated metrics to evaluate recovery capital and motivation, limiting interpretation of the growing psychedelic literature. This commentary describes the current state of this line of investigation and potential impact of psychedelic-assisted psychotherapy on enhancing motivation to change in addiction treatment, and the need for validated metrics to evaluate recovery motivation and capital to assess the potential for psychedelic-assisted psychotherapies to elicit positive, lasting changes in substance use behaviors among those seeking treatment.

KEYWORDS

addiction treatment, substance use disorders, psychedelic-assisted psychotherapy, psychedelics, recovery capital and motivation

Substance use disorders (SUDs) continue to be a source of major morbidity and mortality, accounting for an estimated greater than 20% of all deaths globally (Gakidou et al., 2017). Despite the heavy burden of illness, little is known about how individuals can best overcome and resolve SUDs (Kelly, Bergman, Hoepfner, Vilsaint, & White, 2017). Several evidence-based psychosocial and pharmacological treatments exist; however, they face a range of limitations such as low retention rates, limited availability, and high rates of relapse (BCCSU, 2019). Furthermore, individual differences in neurochemical interactions, psychosocial factors related to substance use (e.g. housing, social security and supports), and understanding of harms associated with ongoing drug use creates a multi-dimensional challenge (Bardo, Neisewander, & Kelly, 2013). Recent data highlight the limited success of currently available addiction treatments in various substance use disorders (Wood, 2018). In addition to this, patient lack of motivation or ambivalence about addiction treatment can serve to inhibit positive treatment outcomes (Palmer, Murphy, Piselli, & Ball, 2009). Consequently, novel evidence-based strategies are urgently needed to enhance support of an individual's internal and external personal, cultural, social, and environmental resources, alongside the development of validated metrics to evaluate recovery motivation in this context.

One important concept that has emerged in the addiction treatment space is that of recovery capital. This refers to quality and quantity of internal and external factors that, if enhanced, may enable an individual with a SUD to remain motivated throughout treatment

*Corresponding author. BC Centre on Substance Use, Department of Medicine, University of British Columbia, 400-1045 Howe Street, Vancouver, BC V6Z 1Y6, Canada. Tel.: +(778) 945-7616; fax: (604) 428-5183.
E-mail: elena.argento@bccsu.ubc.ca

and sustain abstinence (Laudet & White, 2008; White & Cloud, 2008). Treatments that support an individual's recovery capital have the potential to translate into better treatment outcomes (Köpetz, Lejuez, Wiers, & Kruglanski, 2013; Laudet & White, 2008). Motivation is a significant aspect of recovery capital that can be influenced by psychosocial interventions, such as Motivational Enhancement Therapy and Motivational Interviewing (MI) (Miller & Rollnick, 2012; Sellman, Sullivan, Dore, Adamson, & MacEwan, 2001), and is widely acknowledged as a crucial factor in addiction treatment (Miller, 2016). While patients present to treatment settings with a range of internal and external motivations, spanning from relative ambivalence to high motivation for change, approaches that nurture an individual's recovery capital can lead to sustained positive outcomes, including abstinence (Köpetz et al., 2013; Laudet & White, 2008; Palmer et al., 2009). One such re-emerging approach is psychedelic-assisted psychotherapy, which may hold promise in bolstering aspects of recovery capital when used in conjunction with existing treatment modalities (Tupper, Wood, Yensen, & Johnson, 2015).

An increasing body of evidence suggests the adjunctive use of psychedelics (broadly inclusive of psilocybin, lysergic acid diethylamide [LSD], ketamine, and 3,4-Methylenedioxymethamphetamine [MDMA]) with psychotherapy could enhance intrinsic aspects related to recovery capital such as motivation, connectedness, self-efficacy and meaningfulness, subsequently moderating positive SUD outcomes (Argento, Tupper, & Socias, 2019; Bogenschutz et al., 2015; Feduccia et al., 2019; Fuentes, Fonseca, Elices, Farré, & Torrens, 2019; Johnson, Garcia-Romeu, Cosimano, & Griffiths, 2014; Krebs & Johansen, 2012; Krupitsky et al., 2002). A meta-analysis of randomized clinical trials investigating LSD-assisted psychotherapy for AUD showed a pooled benefit difference of 16% in LSD-treated participants that improved their alcohol misuse and 15% that maintained abstinence compared to controls at 6 months follow-up (Krebs & Johansen, 2012), and a growing body of literature now highlights the significant therapeutic utility of LSD in AUD (Fuentes et al., 2019). Likewise, psilocybin-assisted psychotherapy has demonstrated positive changes in abstinence self-efficacy post-treatment and enhancement of motivation for change in AUD (Bogenschutz et al., 2015), yet validated measures of recovery capital remain elusive in the context of this therapeutic approach. Conventional smoking cessation therapies have had limited success, where the three most widely available first-line treatments have observed less than 35% of participants remaining smoke-free after 6 months (Cahill, Stevens, & Lancaster, 2014). Notably, a pilot study investigating psilocybin-assisted psychotherapy for smoking cessation found that 80% of participants, who smoked on average 19 cigarettes a day for an average of 31 years, remained abstinent at 6-months follow-up (Johnson et al., 2014). It is clear that novel, evidence-based approaches are urgently needed for opioid use disorder to address the ongoing overdose crisis. Clients discharged from residential treatment centres have relapse rates of 59% in the first week and 80% within one month (Smyth, Barry, Keenan, &

Ducray, 2010), and retention rates for opioid agonist therapy, considered an evidence-based first-line treatment, are as low as 41% at 12-month follow-up (Nosyk, Marsh, Sun, Schechter, & Anis, 2010). While direct parallels cannot be drawn, historical evidence alongside emergent clinical trials suggest psychedelic-assisted psychotherapy may be beneficial within a multitargeted response to the opioid crisis (Argento et al., 2019). Higher rates of abstinence have been observed among participants with opioid use disorder receiving high dose LSD and ketamine-assisted psychotherapies compared to controls at long term follow ups (Krupitsky et al., 2002; Savage & McCabe, 1973). Although these studies show therapeutic promise of an adjunctive psychedelic approach, they continue to lack validated metrics of quantifying recovery capital, leading to inconsistency and insufficiency of this data in the literature. Given consistent findings from published preliminary studies, larger clinical trials are needed and are now underway in the field to further assess the therapeutic potential of psychedelic-assisted psychotherapy. However, the gap remains for improved metrics to evaluate psychedelic effects (Hendricks, 2018).

Evidence-based psychosocial interventions are increasingly being used to improve treatment outcomes as a monotherapy and in conjunction with other modalities to enhance recovery capital (BCCSU, 2019). For example, Motivational Enhancement Therapy, a structured counseling methodology that employs principles of MI, is designed to explore client attitudes towards change and enhance intrinsic motivation across the spectrum of substance use (Galloway, Polcin, Kielstein, Brown, & Mendelson, 2007; Miller & Rollnick, 2012; Sellman et al., 2001). While MI alone has been shown to have moderate effect sizes for SUDs compared to placebo groups (Burke, Arkowitz, & Menchola, 2003), preliminary research has demonstrated that MI in conjunction with psychedelics may strengthen recovery capital and motivation by generating “mystical-type” experiences that appear to increase self-efficacy post-intervention (Bogenschutz et al., 2015). Psychedelic-occasioned mystical experiences have been shown to enhance ratings of personal meaningfulness, spiritual significance, and connectedness, facilitating deeper personal insight that may lead to lasting behavioral change and positive outlook on life (Carhart-Harris, Erritzoe, Haijen, Kaelen, & Watts, 2018; Griffiths, Richards, McCann, & Jesse, 2006). Indeed, it has been argued that psychedelics might assist in promoting broad positive lifestyle changes, such improved diet, exercise, nature relatedness, and mindfulness practices (Teixeira et al., 2021). This may be partially explained by the meaning enhancing properties of psychedelics (such as mystical experiences and awe) that are thought to have lasting impacts on mood, cognition and substance use behaviors (Garcia-Romeu, Griffiths, & Johnson, 2014; Hendricks, 2018; Johnson, Garcia-Romeu, & Griffiths, 2017). For instance, participants in trials of psychedelic-assisted psychotherapy for SUDs expressed their experiences to be among the five most spiritually significant and personally meaningful ones in their lives (Johnson et al., 2017). Participants articulated

meaningful commitment to change post-intervention and had personalized realizations with increased self-efficacy, highlighting the profound impact psychedelic-assisted psychotherapy may have on recovery capital metrics (Johnson et al., 2017; Nielson, May, Forchimes, & Bogenschütz, 2018; Noorani, Garcia-Romeu, Swift, Griffiths, & Johnson, 2018). The connection between spiritual experiences and eliciting change also raises the possibility for psychedelics to have synergistic effects should they be integrated in broader addiction treatment programs (Yaden et al., 2021). While the literature suggests qualitative changes on recovery capital and motivation are possible, the development of validated quantitative metrics would enhance the ability to formally assess recovery capital.

Structured psychotherapeutic modalities for treating SUD such as MI and Cognitive Behavioural Therapy (CBT), which explore thought patterns and focus on relapse prevention through goal-directed psychotherapy (McHugh, Hearon, & Otto, 2010), have shown modest immediate post-treatment benefits, yet these effects tend to diminish with time (Magill & Ray, 2009). Meanwhile, preliminary studies of psychedelics as adjuncts to psychotherapy have demonstrated positive effects that appear to be effective and sustained (Johnson et al., 2014; Krebs & Johansen, 2012). Recent clinical trials of psychedelic-assisted psychotherapy for treating mental health conditions such as depression, anxiety and posttraumatic stress disorder, which share common neural circuitry and have high comorbidity with addiction (Ly et al., 2018), have demonstrated particularly promising outcomes with lasting effects up to 4.5 years post-intervention (Agin-Liebes et al., 2020; Davis et al., 2020; Mithoefer et al., 2019). As a result, the US Food and Drug Administration has granted psilocybin- and MDMA-assisted psychotherapy “breakthrough therapy designation” for treatment-resistant depression and posttraumatic stress disorder, respectively (Feduccia et al., 2019; Reiff et al., 2020). Thus, the expansion and integration of psychedelic research with innovative psychosocial treatments for SUDs offers a promising modality that may generate longer lasting, internally motivated change and more efficacious treatment outcomes. The validation of metrics to encompass these changes in internal and external motivation would further support the growing body of evidence and provide consistency across future literature.

Capturing and quantifying psychedelic experiences and metrics of recovery capital in the setting of psychedelic-assisted psychotherapy has proven challenging. The notable personal accounts of enhanced sense of connectedness, spirituality, and meaningfulness that individuals experience during psychedelic-assisted psychotherapy play an important role in improving and sustaining treatment outcomes (Johnson et al., 2017; Nielson et al., 2018; Noorani et al., 2018), and yet remain difficult to measure (Maclean, Leoutsakos, Johnson, & Griffiths, 2012). Furthermore, the emotion awe has been hypothesized to be a key mechanism of action underlying mystical experiences in psychedelic-assisted psychotherapy, fostering a sense of connectedness and oneness with others and acting as a principal catalyst for

change within the psychedelic-assisted psychotherapy paradigm (Hendricks, 2018). These factors continue to be difficult to quantify, and although some validated scales for measuring recovery capital exist, such as the Brief Assessment of Recovery Capital (Vilsaint et al., 2017) and Recovery Capital Index (Whitesock, Zhao, Goettsch, & Hanson, 2018), corroborated measures for capturing the unique, intrinsic metrics of recovery capital that appear to be highly influenced by psychedelic-assisted psychotherapy (e.g. motivation, awe, connectedness, mystical experience and spirituality) are lacking. Development and validation of improved measures that have sufficient capacity and sensitivity to evaluate recovery capital metrics in this field will help to bolster the research and evidence emerging from the literature (Carhart-Harris et al., 2018). While psychedelic-assisted psychotherapy may continue to show promise with better validated measures, it is of notable importance that many highly motivated individuals may not be able to achieve their goals due to external structural factors such as housing or employment instability, and stigma (Hennessy, 2017; Laudet & White, 2008). As such, while supporting recovery capital and motivation in addiction settings can help to alleviate stress that comes with treatment and facilitate positive outcomes (Laudet & White, 2008), psychedelic-assisted psychotherapy should be considered alongside broader structural interventions to support individuals in achieving their substance use-related goals.

Though further research is required to substantiate evidence, a growing number of studies, including clinical trials, indicate psychedelic-assisted psychotherapy holds promise when offered concurrently with standard of care interventions. However, there remains an ongoing need in this context to develop and validate new assessment tools that better capture and measure recovery capital and motivation and related factors influenced by psychedelic-assisted psychotherapy that have the potential to sustain long-term benefits including abstinence and remission from SUDs. The global crisis of addiction calls for expanding treatment options beyond the existing therapies for SUDs that are offered today, which are characterized by significant limitations. As the field evolves, formally validated measures for psychedelic effects and impacts on recovery capital and motivation will enhance our understanding of the psychosocial aspects of an individuals’ journey to remission, and the potential for psychedelics to enhance existing treatments. Continuing research and growing evidence of psychedelic-assisted psychotherapy offers a novel and promising approach to initiate and sustain internally motivated change, optimize addiction treatment outcomes and ultimately advance the way we care for those struggling with SUDs.

REFERENCES

- Agin-Liebes, G. I., Malone, T., Yalch, M. M., Mennenga, S. E., Ponté, K. L., Guss, J., et al. (2020). Long-term follow-up of psilocybin-assisted psychotherapy for psychiatric and existential distress in



- patients with life-threatening cancer. *Journal of Psychopharmacology*, 34(2), 155–166.
- Argento, E., Tupper, K. W., & Socias, M. E. (2019). The tripping point: The potential role of psychedelic-assisted therapy in the response to the opioid crisis. *International Journal of Drug Policy*, 66, 80–81.
- Bardo, M. T., Neisewander, J. L., & Kelly, T. H. (2013). Individual differences and social influences on the neurobehavioral pharmacology of abused drugs. *Pharmacological Reviews*, 65(1), 255–290.
- BCCSU (2019). *Provincial guideline for the clinical management of high-risk drinking and alcohol use disorder*. British Columbia Centre on Substance Use.
- Bogenschutz, M. P., Forcehimes, A. A., Pommy, J. A., Wilcox, C. E., Barbosa, P. C., & Strassman, R. J. (2015). Psilocybin-assisted treatment for alcohol dependence: A proof-of-concept study. *Journal of Psychopharmacology*, 29(3), 289–299.
- Burke, B., Arkowitz, H., & Menchola, M. (2003). The efficacy of motivational interviewing: A meta-analysis of controlled clinical trials. *Journal of Consulting and Clinical Psychology*, 71, 843–861.
- Cahill, K., Stevens, S., & Lancaster, T. (2014). Pharmacological treatments for smoking cessation. *JAMA*, 311(2), 193–194.
- Carhart-Harris, R. L., Erritzoe, D., Haijen, E., Kaelen, M., & Watts, R. (2018). Psychedelics and connectedness. *Psychopharmacology (Berl)*, 235(2), 547–550.
- Davis, A. K., Barrett, F. S., May, D. G., Cosimano, M. P., Sepeda, N. D., Johnson, M. W., et al. (2020). Effects of psilocybin-assisted therapy on major depressive disorder: A randomized clinical trial. *JAMA Psychiatry*, 78(5), 481–489.
- Feduccia, A. A., Jerome, L., Yazar-Klosinski, B., Emerson, A., Mithoefer, M. C., & Doblin, R. (2019). Breakthrough for trauma treatment: Safety and efficacy of MDMA-assisted psychotherapy compared to paroxetine and sertraline. *Front Psychiatry*, 10, 650.
- Fuentes, J. J., Fonseca, F., Elices, M., Farré, M., & Torrens, M. (2019). Therapeutic use of LSD in psychiatry: A systematic review of randomized-controlled clinical trials. *Front Psychiatry*, 10, 943.
- Gakidou, E., Afshin, A., Abajobir, A. A., Abate, K. H., Abbafati, C., Abbas, K. M., et al. (2017). Global, regional, and national comparative risk assessment of 84 behavioural, environmental and occupational, and metabolic risks or clusters of risks, 1990–2016: A systematic analysis for the global burden of disease study 2016. *The Lancet*, 390(10100), 1345–1422.
- Galloway, G. P., Polcin, D., Kielstein, A., Brown, M., & Mendelson, J. (2007). A nine session manual of motivational enhancement therapy for methamphetamine dependence: Adherence and efficacy. *Journal of Psychoactive Drugs*, (Suppl 4), 393–400.
- Garcia-Romeu, A., Griffiths, R. R., & Johnson, M. W. (2014). Psilocybin-occasioned mystical experiences in the treatment of tobacco addiction. *Current Drug Abuse Reviews*, 7(3), 157–164.
- Griffiths, R. R., Richards, W. A., McCann, U., & Jesse, R. (2006). Psilocybin can occasion mystical-type experiences having substantial and sustained personal meaning and spiritual significance. *Psychopharmacology (Berl)*, 187(3), 268–283; discussion 284–292.
- Hendricks, P. (2018). *Psilocybin treatment of cocaine use disorder*. Paper Presented at the College on Problems of Drug Dependence 80th Annual Scientific Meeting.
- Hendricks, P. S. (2018). Awe: A putative mechanism underlying the effects of classic psychedelic-assisted psychotherapy. *International Review of Psychiatry*, 30(4), 331–342.
- Hennessy, E. A. (2017). Recovery capital: A systematic review of the literature. *Addiction Research and Theory*, 25(5), 349–360.
- Johnson, M. W., Garcia-Romeu, A., Cosimano, M. P., & Griffiths, R. R. (2014). Pilot study of the 5-HT_{2A}R agonist psilocybin in the treatment of tobacco addiction. *Journal of Psychopharmacology*, 28(11), 983–992.
- Johnson, M. W., Garcia-Romeu, A., & Griffiths, R. R. (2017). Long-term follow-up of psilocybin-facilitated smoking cessation. *The American Journal of Drug and Alcohol Abuse*, 43(1), 55–60.
- Kelly, J. F., Bergman, B., Hoepfner, B. B., Vilsaint, C., & White, W. L. (2017). Prevalence and pathways of recovery from drug and alcohol problems in the United States population: Implications for practice, research, and policy. *Drug and Alcohol Dependence*, 181, 162–169.
- Köpetz, C. E., Lejuez, C. W., Wiers, R. W., & Kruglanski, A. W. (2013). Motivation and self-regulation in addiction: A call for convergence. *Perspectives on Psychological Science*, 8(1), 3–24.
- Krebs, T. S., & Johansen, P. (2012). Lysergic acid diethylamide (LSD) for alcoholism: meta-analysis of randomized controlled trials. *Journal of Psychopharmacology*, 26(7), 994–1002.
- Krupitsky, E., Burakov, A., Romanova, T., Dunaevsky, I., Strassman, R., & Grinenko, A. (2002). Ketamine psychotherapy for heroin addiction: Immediate effects and two-year follow-up. *Journal of Substance Abuse Treatment*, 23(4), 273–283.
- Laudet, A. B., & White, W. L. (2008). Recovery capital as prospective predictor of sustained recovery, life satisfaction, and stress among former poly-substance users. *Substance Use and Misuse*, 43(1), 27–54.
- Ly, C., Greb, A. C., Cameron, L. P., Wong, J. M., Barragan, E. V., Wilson, P. C., et al. (2018). Psychedelics promote structural and functional neural plasticity. *Cell Reports*, 23(11), 3170–3182.
- Macleod, K. A., Leoutsakos, J. M., Johnson, M. W., & Griffiths, R. R. (2012). Factor Analysis of the mystical experience questionnaire: A study of experiences occasioned by the Hallucinogen psilocybin. *Journal for the Scientific Study of Religion*, 51(4), 721–737.
- Magill, M., & Ray, L. A. (2009). Cognitive-behavioral treatment with adult alcohol and illicit drug users: A meta-analysis of randomized controlled trials. *Journal of Studies on Alcohol and Drugs*, 70(4), 516–527.
- McHugh, R. K., Hearon, B. A., & Otto, M. W. (2010). Cognitive behavioral therapy for substance use disorders. *The Psychiatric Clinics of North America*, 33(3), 511–525.
- Miller, W. R. (2016). Sacred cows and greener pastures: Reflections from 40 Years in addiction research. *Alcoholism Treatment Quarterly*, 34(1), 92–115.
- Miller, W. R., & Rollnick, S. (2012). *Motivational interviewing, third edition: Helping people change*. The Guildford Press.
- Mithoefer, M. C., Feduccia, A. A., Jerome, L., Mithoefer, A., Wagner, M., Walsh, Z., et al. (2019). MDMA-assisted psychotherapy for treatment of PTSD: Study design and rationale for phase 3 trials based on pooled analysis of six phase 2



- randomized controlled trials. *Psychopharmacology (Berl)*, 236(9), 2735–2745.
- Nielson, E. M., May, D. G., Forcehimes, A. A., & Bogenschutz, M. P. (2018). The psychedelic debriefing in alcohol dependence treatment: Illustrating key change phenomena through qualitative content analysis of clinical sessions. *Front Pharmacology*, 9, 132.
- Noorani, T., Garcia-Romeu, A., Swift, T. C., Griffiths, R. R., & Johnson, M. W. (2018). Psychedelic therapy for smoking cessation: Qualitative analysis of participant accounts. *Journal of Psychopharmacology*, 32(7), 756–769.
- Nosyk, B., Marsh, D. C., Sun, H., Schechter, M. T., & Anis, A. H. (2010). Trends in methadone maintenance treatment participation, retention, and compliance to dosing guidelines in British Columbia, Canada: 1996–2006. *Journal of Substance Abuse Treatment*, 39(1), 22–31.
- Palmer, R. S., Murphy, M. K., Piselli, A., & Ball, S. A. (2009). Substance user treatment dropout from client and clinician perspectives: A pilot study. *Substance Use and Misuse*, 44(7), 1021–1038.
- Reiff, C. M., Richman, E. E., Nemeroff, C. B., Carpenter, L. L., Widge, A. S., Rodriguez, C. I., et al. (2020). Psychedelics and psychedelic-assisted psychotherapy. *The American Journal of Psychiatry*, 177(5), 391–410.
- Savage, C., & McCabe, O. L. (1973). Residential psychedelic (LSD) therapy for the narcotic addict. A controlled study. *Archives of General Psychiatry*, 28(6), 808–814.
- Sellman, J. D., Sullivan, P. F., Dore, G. M., Adamson, S. J., & MacEwan, I. (2001). A randomized controlled trial of motivational enhancement therapy (MET) for mild to moderate alcohol dependence. *Journal of Studies on Alcohol*, 62(3), 389–396.
- Smyth, B. P., Barry, J., Keenan, E., & Ducray, K. (2010). Lapse and relapse following inpatient treatment of opiate dependence. *Irish Medical Journal*, 103(6), 176–179.
- Teixeira, P. J., Johnson, M. W., Timmermann, C., Watts, R., Erritzoe, D., Douglass, H., et al. (2021). Psychedelics and health behaviour change. *Journal of Psychopharmacology*, 2698811211008554.
- Tupper, K. W., Wood, E., Yensen, R., & Johnson, M. W. (2015). Psychedelic medicine: A re-emerging therapeutic paradigm. *Canadian Medical Association Journal*, 187(14), 1054–1059.
- Vilsaint, C. L., Kelly, J. F., Bergman, B. G., Groshkova, T., Best, D., & White, W. (2017). Development and validation of a Brief Assessment of Recovery Capital (BARC-10) for alcohol and drug use disorder. *Drug and Alcohol Dependence*, 177, 71–76 (1879-0046 (Electronic)).
- White, W., & Cloud, W. (2008). Recovery capital: A primer for addictions professionals. *Counselor*, 9, 22–27.
- Whitesock, D., Zhao, J., Goettsch, K., & Hanson, J. (2018). Validating a survey for addiction wellness: The recovery capital Index. *South Dakota Medicine*, 71(5), 202–212.
- Wood, E. (2018). Strategies for reducing opioid-overdose deaths — lessons from Canada. *New England Journal of Medicine*, 378(17), 1565–1567.
- Yaden, D. B., Berghella, A. P., Regier, P. S., Garcia-Romeu, A., Johnson, M. W., & Hendricks, P. S. (2021). Classic psychedelics in the treatment of substance use disorder: Potential synergies with twelve-step programs. *International Journal of Drug Policy*, 98, 103380.

